Ultimate Nutrition
Tonalin® CLA

For many years, performance nutrition experts essentially dismissed fats, assuming they didn't play a useful role in exercise. Experts focused instead on the protein-sparing and energy-producing effects of carbohydrates and how protein/amino acids affect nitrogen balance. However, it is now well known that fatty acids are not only essential for proper health, but many have additional benefits for health and fitness. Conjugated linoleic acid (CLA) is a specialized fat that has been gaining popularity in the athletic community. Current research studies have shown a positive change in body composition among athletes after supplementing with CLA.[1-5]

CLA is a slightly altered form (i.e., isomer) of linoleic acid, an omega-6 fatty acid, essential to human health. CLA is a naturally occurring fatty acid that cannot be produced by the human body but can be obtained in small amounts through foods such as dairy products (e.g., cheese and whole milk), butter, and beef.[6] The typical North American diet does not provide enough CLA to get the related benefits, and the foods that do contain small amounts typically have high amounts of saturated fat. As the research started to show positive benefits from CLA supplementation, a more economical and efficient way to provide CLA was required. Fortunately, scientists are able to convert the linoleic acid of pure safflower oil into CLA. Advanced lipid technology provided precise CLA intake through supplementation without high caloric food consumption and gave rise to Tonalin® CLA. Tonalin® CLA is produced from linoleic acid found in natural safflower oil utilizing a process that gently converts the linoleic acid into CLA.

The reported effects of CLA on body composition, specifically a reduction in body fat mass and a preservation of lean body mass, generated interest among the athletic community. While the mechanism by which CLA works is still an active area of research, it is known that CLA can be readily metabolized in human tissue. Current research shows that scientists have determined several potential mechanisms of action. Research suggests that CLA may induce adipocyte (i.e., fat cell) apoptosis (i.e., programmed cell death), resulting in fewer fat cells.[7] CLA has been shown to inhibit lipoprotein lipase, an enzyme which helps break down fats so that it can be absorbed and stored as fat cells ,and increase the carnitine palmitoyltransferase enzyme that transfers fat into the muscle cells to be burned. These interactions result in slowing the amount of fat that is deposited and stored in the body along with an increase in fat oxidation.[8] CLA has also been shown to modify energy expenditure/metabolic rate and bind to a receptor in fat tissue, thereby preventing the accumulation of triglycerides into fat cells.[9-12] Although the mechanisms by which CLA exerts its action on body weight have not been fully elucidated, it seems clear that CLA is involved in altering body composition in humans by reducing body fat. Clinical studies have confirmed that CLA reduces body fat and weight in various ways from decreasing the amount of fat your body stores to increasing the rate of fat metabolism/breakdown to preserving and sometimes increasing muscle mass.[2]

A considerable amount of research has used CLA for its ability to modify body composition.[1,2,13,14] In one short-term study, CLA users who did not make changes in their diet and lifestyle experienced a decrease in fat.[5] Longer term studies in men and women supplementing with CLA have shown similar results with reductions in body fat mass while maintaining lean body mass.[1,2,15] Some subjects supplementing with CLA have also experienced increases in lean body mass.[4] However, these subjects also increased the intensity of their training. It appears that individuals experience a reduction in fat mass and increased lean mass when CLA is used in conjunction with a resistance program.[13] Studies also suggest that the maintenance of and/or increase of lean body mass contributes to a more efficient metabolism, helping to maintain energy expenditure.[2] This is a rational explanation because it is well known that muscle burns more calories than fat.

There is no quick and easy way to lose weight, but taking Tonalin® CLA from Ultimate Nutrition® in combination with a healthy, well-balanced diet and a regular fitness regimen can help reduce stubborn fat. Tonalin® CLA offers an interesting approach to supplementation that can be taken daily as a successful aid to healthy weight loss and fat reduction. Although the optimal dose is unknown, a minimum dosage of 1.8 grams of CLA/day has been shown to achieve a slight reduction in body fat.[14] Additional studies have confirmed that CLA can help reduce unwanted body fat and preserve or even increase lean muscle mass when combined with exercise. Therefore, research proves that Tonalin® CLA can be a valuable addition which aids in fat loss when combined with regular exercise and a healthy diet.

FAQs

Q: How should I take Tonalin CLA?
A: As an adult dietary supplement, take one softgel (1000 mg) with each meal.

Q: Are there any side effects associated with CLA?
A: Research has shown CLA is safe to take and free of any known side effects when consumed according to the recommended dosage. Numerous clinical studies have been conducted on the safety of CLA and no adverse side effects have been reported.

Q: Can I use CLA if I'm taking prescription drugs?
A: It is advised you consult with your doctor or Health Care provider before starting any supplement if you are taking any medications or prescription drugs.

Q: Is there anything I should avoid combining with CLA?
A: Do not take chitosan and CLA together. The chitosan will absorb the CLA and prevent it from getting into the bloodstream.

REFERENCES

1. Gaullier et al. (2004). Conjugated linoleic acid (CLA) supplementation for one year reduces body fat mass in healthy, overweight humans. Am J Clin Nutr 79, 1118–1125.
2. Gaullier et al. (2005). Supplementation with Conjugated Linoleic Acid for 24 months is well tolerated by and reduces body fat mass in healthy, overweight humans. J Nutr. 135:778-784
3. Gaullier et al. (2007). Six months supplementation with conjugated linoleic acid induces regional-specific fat mass decreases in overweight and obese. Br J Nutr. 97:550-560.
4. Berven et al. (2000). Safety of conjugated linoleic acid in overweight or obese human volunteers. Eur J Lipid Sci Technol. 102:455-462.
5. Riserus et al. (2001). Conjugated Linoleic Acid reduces abdominal fat in overweight adults. International Journal of Obesity. 25(8): 1129-1135
6. Silveria et al. (2007). Conjugated linoleic acid (CLA) and obesity. Public Health Nutrition. 10(10A):1181-1186
7. Evans M et al. (2000) Conjugated linoleic acid suppresses triglyceride accumulation and induces apoptosis in 3T3-L1 preadipocytes. Lipids 35, 899–910.

8. Park Y & Pariza M (2001) Lipoxygenase inhibitors inhibit heparinreleasable lipoprotein lipase activity in 3T3-L1 adipocytes and enhance body fat reduction in mice by conjugated linoleic acid. Biochim Biophys Acta 1534, 27–33.
9. West D et al. (1998) Effects of conjugated linoleic acid on body fat and energy metabolism in the mouse. Am J Physiol 275, R667–R672.
10. Terpstra A (2001) Differences between humans and mice in efficacy of the body fat lowering effect of conjugated linoleic acid: role of metabolic rate. J Nutr 131, 2067–2068.
11. Kallen CB & Lazar MA (1996) Antidiabetic thiazolidinediones inhibit leptin ob gene expression in 3T3-L1 adipocytes. Proc Natl Acad Sci USA 93, 5793–5796.
12. Granlud L et al. (2003) Trans 10, cis 12 conjugated linoleic acid prevents triacylglycerol accumulation in adipocytes by acting as a PPARgamma modulator. J Lipid Res 44, 1441–1452.
13. Blankson et al. (2000). Conjugated linoleic acid reduces body fat mass in overweight and obese humans. J. Nutr 130: 2943-2948.
14. Thom et al. (2001). Conjugated linoleic acid reduces body fat in healthy exercising humans. J Int Med Res. 29:392-296.
15. Mougios et al. (2001). Effect of supplementation with conjugated linoleic acid on human serum lipids and body fat. J Nutr Biochem. 12:585-594